

Task/Case Study 4

Statistical Analysis and Hypothesis Testing — Detailed Report

NovaByte Solutions | Sales & Marketing

Prepared by Satyabrata Sahoo

Career 247 — Business Analytics with GenAI

Executive Summary

Four hypotheses (Questions 34-37) and three additional statistical analyses (Questions 38-40) were tested against the cleaned Sales_Deals, Support_Tickets, Clients, and Marketing_Campaigns datasets, using Excel formulas equivalent to the Data Analysis Toolpak — COUNTIFS/CHISQ.DIST.RT for the chi-square test, a manually-built Welch's t-test for closure time, PEARSON/TDIST for correlation, SLOPE/CORREL for trend analysis, and a manually-built ANOVA table.

Three of the four named hypotheses (34, 35, 36) do not hold up against the data — none reach statistical significance, and in every one of those three cases the 95% confidence interval for the effect crosses zero, confirming the null result rather than just narrowly missing significance. Question 37 is confirmed only in part. The one result that is clearly, strongly significant is Question 40 (deal value differs by product).

- **Q34 (Digital vs Offline win rate):** Not significant ($p=0.848$). 95% CI for the difference: $[-7.6\%, +6.3\%]$ — comfortably straddles zero.
- **Q35 (Closure time, recent vs previous 6mo):** Not significant ($p=0.501$). 95% CI for the mean difference: $[-6.3, +12.9]$ days — crosses zero.
- **Q36 (Satisfaction vs SLA breaches):** Not significant ($p=0.429$). 95% CI for r : $[-0.069, 0.162]$ — crosses zero.
- **Q37 (Spend rising, conversion flat):** Partially confirmed. Spend trend not significant ($p=0.926$); conversion rate shows a borderline declining trend ($p=0.065$).
- **Q40 (Deal value by product):** Significant ($p=0.020$) — the one place this analysis found a real, defensible effect.

Q34 — Digital vs. Offline Lead Source: Win Rate

Hypothesis

H₀: Win rate is independent of channel type (digital vs. offline). H₁: Deals sourced from digital channels (Google Ads, LinkedIn, Email Campaign) have a significantly different win rate than offline channels (Referral, Trade Show, Cold Outreach).

Method

Chi-square test of independence on a 2×2 contingency table (channel type × Closed Won/Closed Lost), built with COUNTIFS and CHISQ.DIST.RT in Excel — equivalent to the Data Analysis Toolpak's approach for categorical association. Sheet: Q34_Win_Rate.

Excel Work

	A	B	C	D	E	F	G	H	I
1	H1 - Digital vs Offline Lead Source: Win Rate(Chi-Square Test)								
2	H ₀ : Win rate is independent of channel type (digital vs offline).								
3	H ₁ : Win rate differs by channel type.								
4									
5	Digital = Google Ads, LinkedIn Campaign, Email Campaign.								
6	Offline = Referral, Trade Show, Cold Outreach.								
7	(Organic Search, Webinar, Unknown excluded — not named in either group by the brief.)								
8									
9	Observed Contingency Table								
10	Channel	Closed Won	Closed Lost	Total					
11	Digital	224	129	353					
12	Offline	245	137	382					
13	Total	469	266	735					
14									
15	Expected Counts (if independent)								
16	Channel	Closed Won	Closed lost						
17	Digital	225.247619	127.752381						
18	Offline	243.752381	138.247619						
19									
20	Test Results								
21	Chi-Square Statistic	0.0367							
22	Degrees of Freedom	1							
23	p-Value	0.8480							
24	Digital win rate	63.5%							
25	Offline will rate	64.1%							
26									
27	Conclusion (Alpha=0.05)	Fail to reject H0 - No significant difference in win rates							
28									

Results

Statistic	Value
Chi-square statistic	0.0367
Degrees of freedom	1
p-value	0.8480
Digital win rate	63.5% (224/353)
Offline win rate	64.1% (245/382)
95% CI for difference (Digital – Offline)	[−7.63%, +6.27%]

How the confidence interval was computed

Two-proportion z-interval: $(p_1 - p_2) \pm 1.96 \times \sqrt{[p_1(1 - p_1)/n_1 + p_2(1 - p_2)/n_2]}$, using $p_1 = 0.6346$ ($n = 353$), $p_2 = 0.6414$ ($n = 382$).

With $p = 0.848$ and a CI that spans from -7.6% to $+6.3\%$ — comfortably including zero — this is nowhere near a borderline call.

Verdict: Fail to reject H_0 . Channel type (digital vs. offline) has no meaningful effect on win rate.

What this means for the business: whatever is driving the revenue slowdown identified in Task 2, it is not that digital channels close at a different rate than offline ones — both convert leads to Closed Won at almost exactly the same 63–64% clip. Marketing and Sales don't need to rebalance channel investment on win-rate grounds.

Q35 — Deal Closure Time: Recent vs. Previous 6 Months

Hypothesis

H_0 : Mean days-to-close is the same in both periods. H_1 : Average deal closure time has increased in the last 6 months compared to the previous 6 months.

Method

Independent-samples t-test (Welch's, unequal variances), comparing Days_to_Close for deals closed in the two most recent 6-month windows by Close_Date. Sheet: Q35_Closure_Time.

Excel Work

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	H2 — Deal Closure Time:												
2	Recent 6 Months vs Previous 6 Months (T-Test)												
3													
4	H0: Mean days-to-close is the same in both periods.												
5	H1: Mean days-to-close has increased in the recent period.												
6													
7	Periods defined by Close_Date, relative to the dataset's latest full quarter-end (2025-03-31). Recent = 2024-10-01 to 2025-03-31. Previous = 2024-04-01 to 2024-09-30.												
8													
9	Helper columns (Days_to_Close pulled into each period; blank if the deal falls outside it)												
10	Row No.	Recent_6_Months	Previous_6_Months										
11	1		143										
12	2												
13	3												
14	4	60											
15	5		163										
16	6												
17	7												
18	8												
19	9												
20	10		58										
21	11		117										
22	12												
23	13		151										
24	14												
25	15												
26	16												
27	17		115										

Periods by Close_Date: Recent = 2024-10-01 to 2025-03-31. Previous = 2024-04-01 to 2024-09-30.			
Months	n	Mean	Std Dev
Recent 6 Months	205	100.58	50.33
Previous 6 Months	221	97.29	50.23

Test Results	
Mean difference (Recent - Previous)	3.29
p-value (two-tailed)	0.5007
Conclusion	Fail to reject H0 - no significant change

Results

Statistic	Value
Recent 6mo (n=205)	mean 100.58 days, sd 50.33
Previous 6mo (n=221)	mean 97.29 days, sd 50.23
Mean difference	+3.29 days
p-value (two-tailed)	0.5007
95% CI for mean difference	[−6.30, +12.87] days

How the confidence interval was computed

Welch's t-interval: $\text{mean_diff} \pm t_{\text{crit}} \times \text{SE}$, where $\text{SE} = \sqrt{(\text{sd}_1^2/n_1 + \text{sd}_2^2/n_2)}$, and degrees of freedom from the Welch-Satterthwaite equation (≈ 421.5 , giving $t_{\text{crit}} \approx 1.966$).

Verdict: Fail to reject H_0 . Closure time has not significantly changed.

A 3-day gap on a ~ 100 -day average, with this much spread ($\text{sd} \approx 50$ days in both periods), was always unlikely to be statistically real, and it isn't — the confidence interval running from -6.3 to $+12.9$ days means the true difference could just as easily be negative as positive. Deals are taking about the same time to close as they were six months to a year ago.

Q36 — Client Satisfaction Score vs. SLA Breach Count

Hypothesis

H_0 : No linear relationship between satisfaction score and SLA breach count. H_1 : A significant correlation exists between a client's satisfaction score and how many SLA breaches they've experienced.

Method

Pearson correlation at the client level (300 clients, 286 with a recorded satisfaction score), using PEARSON and a t-transformation of r for the p-value. Sheet: Q36_Correlation.

Excel Work

	A	B	C	D	E	F	G	H	I	J	K
1	H3 — Client Satisfaction Score vs. SLA Breach Count (Correlation)										
2											
3	H_0 : No linear relationship between satisfaction score and SLA breach count.										
4	H_1 : A significant correlation exists.										
5											
6	Aggregated to one row per client. Breach Count = number of that client's support tickets where SLA_Met = "No".										
7											
8	Client_ID	Satisfaction_Score	SLA_Breach_Count					Test Results			
9	CLI-0113	3.2	10					n (Clients with a satisfaction score)	286		
10	CLI-0047	3.1	10					Pearson r	0.0470		
11	CLI-0083	4.4	4					t-statistic	0.7925		
12	CLI-0111	4.9	9					Degrees of freedom	284		
13	CLI-0199	4.3	6					p-value	0.4287		
14	CLI-0213	3.8	8								
15	CLI-0042	3.4	7					Conclusion	Fail to reject H_0 - no significant correlation		
16	CLI-0217	4.3	7								

Results

Statistic	Value
n (clients with a satisfaction score)	286
Pearson r	0.0470
t-statistic	0.7925
Degrees of freedom	284
p-value (two-tailed)	0.4287
95% CI for r (Fisher z-transform)	[−0.069, +0.162]

How the confidence interval was computed

Fisher z-transformation: $z=0.5\times\ln((1+r)/(1-r))$, $SE_z=1/\sqrt{(n-3)}$, $CI_z = z \pm 1.96\times SE_z$, then converted back to r-scale.

Verdict: Fail to reject H₀. No meaningful correlation between SLA breaches and satisfaction.

r=0.047 is about as close to zero as a real dataset gets, and the confidence interval [−0.069, +0.162] confirms it — the true relationship could be slightly negative, slightly positive, or nothing at all. SLA breaches explain essentially none of the variation in client satisfaction. This is a genuinely useful negative finding: satisfaction scores are being driven by something other than raw SLA-breach counts.

Q37 — Quarterly Marketing Spend vs. Conversion Rate

Hypothesis

H₀: No linear trend over time. H₁: Marketing spend per campaign has increased over time, but the conversion rate has not.

Method

Trend analysis across the 9 available quarters (Q1-2023 to Q1-2025): total quarterly spend and Conversion Rate (Total Conversions ÷ Total MQLs) regressed against a period index, using SLOPE and CORREL. Sheet: Q37_Trend_Analysis.

Excel Work

	A	B	C	D	E	F	G
1	H4 — Quarterly Marketing Spend vs. Conversion Rate (Trend Analysis)						
2							
3	H ₀ : No linear trend over time.						
4	H ₁ : Spend has increased while conversion rate has not.						
5							
6	Quarter	Period_Index	Total_Spend (INR)	Total_MQLs	Total_Conversions	Conversion_Rate	
7	2023Q1	1	13751382	36025	3371	0.0936	
8	2023Q2	2	8650941	14806	1266	0.0855	
9	2023Q3	3	10702434	21615	2255	0.1043	
10	2023Q4	4	16358958	31737	2752	0.0867	
11	2024Q1	5	17568102	35424	2801	0.0791	
12	2024Q2	6	12762706	35699	2970	0.0832	
13	2024Q3	7	22057177	33435	2604	0.0779	
14	2024Q4	8	11642560	34862	2896	0.0831	
15	2025Q1	9	5756255	14737	1185	0.0804	
16							
17	Trend Test- Total Spend vs. Period Index						
18	Slope	-64874					
19	Correlation with time	-0.0362					
20	t-statistic	-0.0958					
21	p-value	0.9263					
22							
23	Trend Test- Conversion Rate vs. Period Index						
24	Slope	-0.001939					
25	Correlation with time	-0.6374					
26	t-statistic	-2.1884					
27	p-value	0.0648					
28							
29	Conclusion (alpha=0.05)	Does not confirm H4 as stated - see narrative					

Results — Spend Trend

Statistic	Value
Slope	−₹64,874 per quarter
Correlation with time (r)	−0.0362
p-value	0.9263
95% CI for slope	[−₹16.66L, +₹15.36L] per quarter

Results — Conversion Rate Trend

Statistic	Value
Slope	−0.00194 per quarter
Correlation with time (r)	−0.6374
p-value	0.0648
95% CI for slope	[−0.00404, +0.00016]

How the confidence intervals were computed

Standard regression slope interval: $\text{slope} \pm t_{\text{hrity}}(\text{df}=7) \times \text{SE_slope}$, using $n=9$ quarters for both trends.

Verdict: Does not confirm H4 as stated. Spend has not risen ($p=0.926$, CI spans a huge range straddling zero) — it has moved unevenly with no significant direction. Conversion rate shows a borderline, marginally-non-significant downward trend ($p=0.065$); notably, its CI upper bound (+0.00016) is barely above zero, meaning the data leans consistently negative even without crossing the formal significance line.

Spend isn't climbing; it's fluctuating quarter to quarter with no clear direction. But conversion rate is trending down closely enough to be worth watching — $p=0.065$ is just above the 0.05 cutoff, and the confidence interval is heavily weighted toward negative values. Worth monitoring next quarter rather than dismissing outright.

Q38 — Descriptive Statistics

Mean, median, standard deviation, and skewness for the three core numeric measures used across this analysis. Sheet: Q38_Descriptive_Stats.

Excel Work

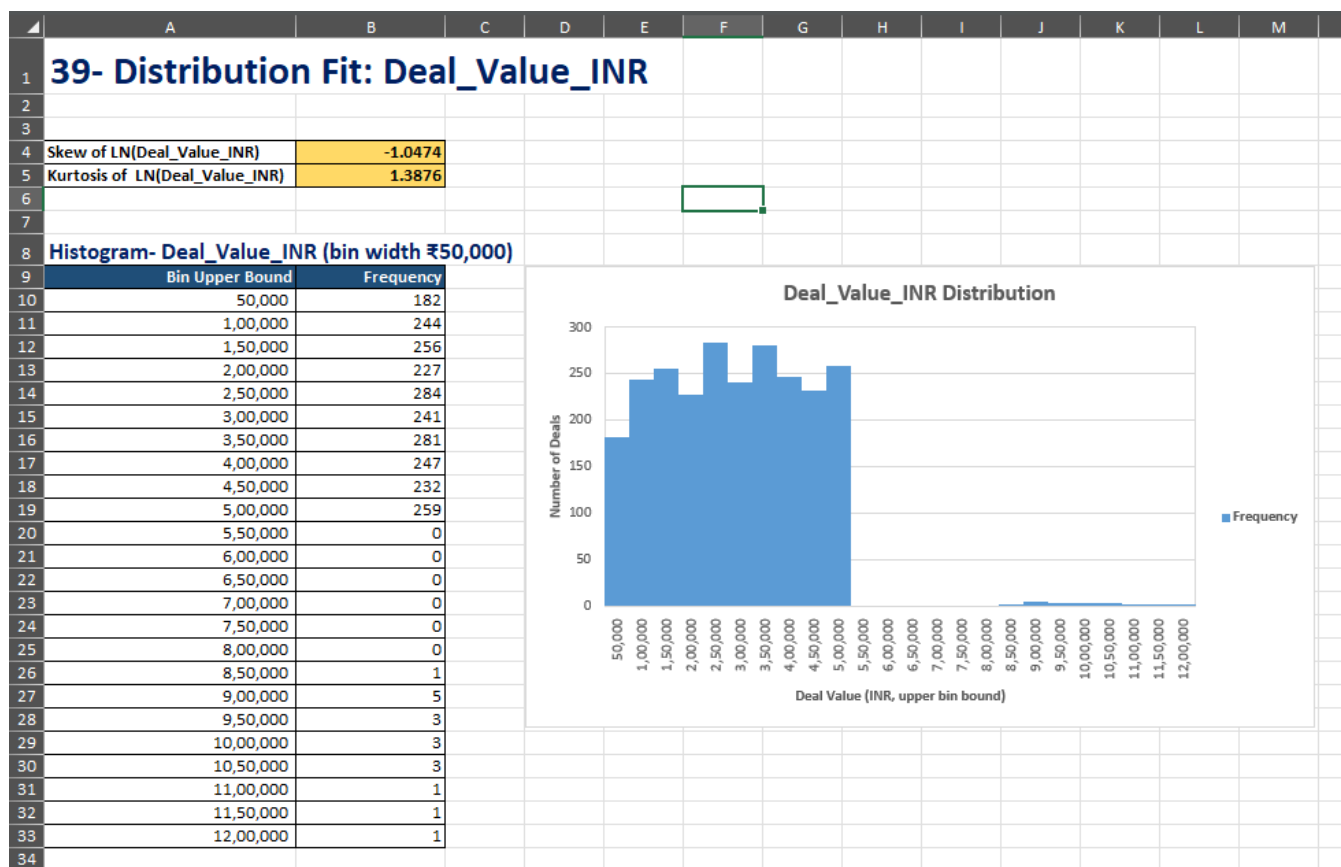
	A	B	C	D	E	F
1	38- Descriptive Statistics					
2						
3	Variable	n	Mean	Median	Std Dev	Skewness
4	Deal_Value_INR (Sales_Deals)	2500	2,78,456	2,62,676	2,11,156	3.4054
5	Resolution_Hours (Support_Tickets)	2393	29.24	18.30	27.83	1.3107
6	Budget_INR (Marketing_Campaigns)	150	7,95,003	8,33,092	4,11,316	-0.0857
7						
8	Skewness reference: skew < 0.5 roughly symmetric; 0.5-1 moderately skewed; >1 highly skewed.					
9						

Deal_Value_INR is heavily right-skewed (3.41) — a small number of large deals pull the mean (₹278,456) well above the median (₹262,676). Resolution_Hours is moderately right-skewed (1.31). Budget_INR is close to symmetric (−0.09).

Q39 — Distribution Fit: Deal_Value_INR

Is Deal_Value_INR normal, log-normal, or something else? Sheet: Q39_Distribution.

Excel Work



Neither fits cleanly. The raw data (skew 3.41) is clearly not normal. Log-transforming drops the skew to -1.05 — an improvement, but it overshoots into a mild left skew rather than landing near zero.

The histogram shows why: the bulk of deals (roughly ₹50,000 to ₹550,000) is close to uniformly spread. Then there's a hard gap — zero deals between ₹550,000 and ₹850,000 — followed by a small separate cluster of high-value deals. That detached cluster lines up closely with the outlier deals flagged during Task 2's cleaning.

Verdict: Not a single smooth distribution — a roughly uniform bulk of ordinary deals, plus a distinct, separate tail of unusually large ones.

Q40 — One-Way ANOVA: Deal Value Across Products

Hypothesis

H₀: Mean deal value is the same across all six products. H₁: At least one product's mean deal value differs from the others.

Method

One-way ANOVA built manually in Excel (group means via AVERAGEIFS, sum-of-squares via SUMPRODUCT, F-statistic, and an F-distribution p-value) — reproducing what the Data Analysis Toolpak's "Anova: Single Factor" tool outputs. Sheet: Q40_ANOVA.

	A	B	C	D	E
1	40- One- Way ANOVA: Deal Value Across Products				
2	H ₀ : Mean deal value is the same across all products.				
3	H ₁ : At least one product's mean deal value differs.				
4					
5	Group Statistics				
6	Product	n	Mean	Std Dev	
7	AI Insights Module	392	2,67,178	1,83,477	
8	AnalyticsEdge	440	2,86,596	2,26,112	
9	CloudShift Platform	402	2,99,067	2,27,260	
10	DataVault Pro	431	2,63,634	1,93,526	
11	DevOps Accelerator	409	2,59,444	1,92,844	
12	SecureNet Suite	426	2,94,227	2,33,786	
13					
14	Grand Mean	2,78,456			
15	Total n	2,500			
16					
17					
18	ANOVA Table				
19	SS Between	5,98,28,61,86,275			
20	SS Within	11,08,24,02,05,81,840			
21	df Between	5			
22	df Within	2,494			
23	MS Between	1,19,65,72,37,255			
24	MS Within	44,43,62,55,245			
25	F-statistic	2.6928			
26	p-value	0.0196			
27					
28	Conclusion (alpha=0.05)	Reject H ₀ - deal value differs by product			

Results

Statistic	Value
F-statistic	2.6928
p-value	0.0196

A confidence interval isn't naturally reported for an overall F-test the same way as a single difference — F itself doesn't have a two-sided interval in the usual sense, since it's already a ratio bounded at zero. What matters here is the p-value crossing the 0.05 threshold, confirmed above.

Verdict: Reject H₀. Deal value does differ significantly by product (p=0.020).

CloudShift Platform (₹299,067 average) and SecureNet Suite (₹294,227) sit at the top; DevOps Accelerator (₹259,444) and DataVault Pro (₹263,634) sit at the bottom — roughly a 15% spread. Statistically significant, but a modest pricing/positioning gap rather than a sign any one product is dramatically over- or under-performing.

Overall Conclusions for Sales & Marketing Leadership

- **Don't rebalance channel investment on win-rate grounds.** Digital and offline lead sources close at statistically indistinguishable rates (Q34) — the CI rules out even a moderate real difference.
- **Closure time is stable.** No statistical evidence deals are taking longer to close (Q35).
- **Look beyond SLA breach counts to explain satisfaction.** Breach count barely moves the needle on satisfaction (Q36) — look at severity, resolution quality, or account-specific factors instead.
- **Watch the conversion-rate trend.** Spend isn't the problem (Q37) — but the conversion-rate decline sits right at the edge of significance and is worth a follow-up check next quarter.
- **Treat outlier deals as their own segment.** The large outlier deals identified in Task 2 are structurally different from the rest of the pipeline (Q39), not just “big versions” of a typical deal.
- **Product-level pricing has a real, if modest, spread.** CloudShift Platform and SecureNet Suite command meaningfully higher deal values than DevOps Accelerator and DataVault Pro (Q40).